Perfect Abstention Is Zero Knowledge: Two Stable Fixed Points of a Self-Referential Abstention Objective in a Small Language Model with Versioned Memory

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Abstract

A hallucination detector should predict its own model's errors, so the natural training target is self-referential: label each example by whether the model *would be wrong if it answered.* We train that target in a 863,730-parameter language model with a co-trained versioned archive, holding architecture, data and budget fixed, and find that the objective has **two stable fixed points**. Nine units split 5–4 between them, and the split is governed by **how much retrieval the model already had when abstention training began**: six units that started from scratch split 2–4, while three that inherited 12,000 steps of retrieval pre-training (RECUP 0.77–0.80) reached the useful point 3 of 3.

The degenerate one is the finding. Four of nine units converge to a state that scores `nose` = 1.0000 and `invento` = 0.0000—*perfect* on both metrics the abstention literature reports—while their global accuracy sits at 0.4065, which is exactly the trivial floor obtained by abstaining on everything. They are not detectors that fail. They are **perfectly calibrated detectors of a generator that knows nothing**, and the calibration is honest: the model does not recover, and it correctly reports that it does not. Their binary head has collapsed to a constant with AUC 0.522–0.578 on its own target, and that constant sits where the prior predicts it: for the youngest unit, logit of the base error rate is 1.513 against a measured median of 1.507.

The bifurcation is decided early and is sharply legible. At step 2500, on the 512-sample evaluation batch the training loop already computes, the five surviving units had emitted 2, 4, 4, 8 and 1 answers; the four degenerate ones had emitted **zero**. The rule "zero versus non-zero at step 2500" separates **40 of 40** runs with a binary head across twelve families in this repository, and one unit survived on a *single* sample out of 512. The early mute phase itself belongs to the self-referential target: on the paired control that shares seed *and* initialization, three units with a fixed, model-independent target read 0.06–0.20 against 0.98–0.99 for their self-referential twins.

The degenerate point is **absorbing**, not merely slow. Measuring recovery inside one unit with paired samples across 4000 further steps gives -0.0021, or 0.4σ: it does not move. We also report a correction to our own first reading, in which a between-unit trend that survives a noise control ($r = 0.986$) was nonetheless not extrapolable, because the curve is concave and a linear fit hid a saturation.

The practical consequence is a metric claim, not an architectural one: **any success criterion that reads abstention rate and fabrication rate without reading global accuracy is maximized by a model that has learned nothing**, and a self-referential objective supplies a training dynamic that reaches that maximum on its own, in a substantial fraction of runs, with no warning from either metric.

Keywords: abstention; hallucination detection; selective prediction; self-referential objectives; training dynamics; degenerate solutions; small language models; persistent memory

1 The metric a silent model wins

The literature on abstention reports two numbers. One is the rate at which a model correctly declines to answer questions it cannot answer; the other is the rate at which it fabricates. Both are desirable, both are bounded, and a system that improves on both is normally taken to have improved.

A model that abstains on every question scores perfectly on both.

This is not a subtle observation, and it is not new as an abstract point: the selective-prediction literature has always paired risk with coverage precisely because risk alone is trivially minimized (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). What we report is that the degenerate corner is not only reachable in principle but is an **attractor of a natural training objective**, that a substantial fraction of runs land in it, and that once landed the state is stable under further training.

The objective in question is the one a hallucination detector actually wants. A detector is useful if it predicts *this model's* errors, so the target should not be "does the answer exist?" but "will this model be wrong if it answers?". That target is available for free during training: take the model's own `argmax`, compare it against the label, and use the comparison as a binary target with the gradient stopped. It is the natural thing to do and, in this bench, it is what produced the units this paper is about.

Its defect is structural rather than incidental. The label is a function of the model's own state, so it moves as the model learns, and the map has more than one consistent solution. At initialization the model is wrong on nearly everything, so the target is the constant 1 and the head learns to predict "I will be wrong"—which is *true*. Escaping requires the generator to start being right before the head has finished settling on the constant. It is a race between two learning timescales, and §5.1 shows that both the seed and the amount of retrieval the model already has when the race starts decide which one wins.

Both outcomes are self-consistent, and that is what makes the degenerate one stable:

fixed point	what the model asserts, and whether it is true
useful	"I retrieve well, and I know when I will fail" — true
degenerate	"I retrieve almost nothing, and I know that I retrieve almost nothing" — **also true**

The degenerate model is not miscalibrated. It is honest, useless, and invisible to the two metrics the field reports.

Contributions.

1. We characterize the degenerate fixed point of a self-referential abstention objective and show it is reached by 4 of the 6 units trained from scratch, and by none of the 3 that inherited retrieval pre-training (§5.1).

2. We show the degenerate units are *not* broken detectors: their heads have collapsed to the prior, and we verify this quantitatively against the closed-form constant, not just qualitatively (§5.2).

3. We show the bifurcation is decided by step 2500 and is separable by a zero/non-zero rule on data the training loop already produces, validated on 40 runs (§5.3).

4. We show the degenerate point is absorbing rather than slow, using the first within-unit measurement of retrieval in this line of work (§5.5).

5. We correct a first reading of our own data in which a real between-unit trend was extrapolated with the wrong functional form (§5.6).

2 Related work

Selective prediction. The risk–coverage framing (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017) exists because abstention trades against usefulness, and SelectiveNet (Geifman and El-Yaniv, 2019) makes the trade explicit by training a selection head against a coverage constraint. Our degenerate units are the coverage constraint's reason for existing, observed as a training outcome rather than as an argument.

Learning to say "I don't know". Adding an `[IDK]` token to the vocabulary (Cohen et al., 2024) and adding a separate binary head are the two dominant interfaces; we compared four of them under matched budget in prior work (Speranza, 2026) and found the head superior in this regime. That comparison used a *fixed* target—"does the answer exist?"— and none of its units entered the state described here. The present paper is what happens when the target is made self-referential instead.

Self-referential and bootstrapped targets. Training a predictor against labels derived from the current model is standard in self-training and pseudo-labelling, and the confirmation-bias failure mode is well documented (Arazo et al., 2020). Our contribution is not that self-referential targets can collapse, which is expected, but the specific shape of the collapse in the abstention setting: the collapsed state is *correct*, scores perfectly on the standard metrics, and is therefore not detectable as a failure by them.

Degenerate solutions and collapse. Representation-learning methods that predict their own outputs collapse to constants without an explicit remedy (Grill et al., 2020; Chen and He, 2021). The mechanism here is the same family—a target that the model can satisfy by moving the target—but the collapsed solution is not a constant representation, it is a constant *decision*, and its constancy is exactly what makes it score well.

3 The bench

The bench is the one described in Speranza (2026); we restate what is needed to read the results and refer there for the rest.

Model. A language model trained *from scratch*, no pretrained component: 863,730 parameters, 3.5 MB, closed vocabulary of 242 human-readable tokens. A delta rule recurrence plus a **co-trained persistent archive** with a learned order stamp in the key, read by softmax and injected at block 0. The archive persists across sessions while the recurrent state is reset between them.

Task. Facts are stated in one session, corrected in another, and queried in a third. The answer is a **single token**, so the metric is exact and deterministic: no LLM judge, no interpretive parser. 40% of questions have no answer ($p_{\text{nose}} = 0.4$); the empirical fraction on the evaluation sample is 0.4065.

Interface. All units in this paper use the separate binary head, which the prior comparison selected: a scalar a projected from the same final state that feeds the vocabulary softmax, +129 parameters. The model abstains when $a > 0$. Reporting the collapse on the *winning* interface is deliberate—it is not an artifact of a weak abstention mechanism.

Metrics. `vigente` is accuracy on the current value *as emitted*, i.e. after the head's decision; `nose` is the abstention rate on questions without an answer; `falsa_abst` is the abstention rate on questions that do have one; `invento` is the rate of answers naming content absent from the archive. We add two instruments:

- **Global accuracy** = (correct answers + correct abstentions)/n. Its trivial floor on this bench is **0.4065**, obtained by abstaining on everything.
- **RECUP**, retrieval accuracy *ignoring the head*: the `argmax` over values on questions that do have an answer, with the abstention decision not applied. `vigente` is a composite of retrieval and decision and is 0.0000 whenever $a > 0$ everywhere, regardless of what the model retrieved; RECUP separates the two.

Measurement protocol. RECUP is measured with $n = 8000$ and a fixed data seed (54321) across every checkpoint, so comparisons are **paired**: the questions are identical and only the model differs. This matters. Measured across six data seeds at $n = 2000$, RECUP has $\sigma = 0.0135$, so an unpaired difference carries $\sigma \approx 0.0191$—larger than the effects of interest. We report p_{flip}, the fraction of questions on which two checkpoints disagree, and derive the paired standard deviation from it rather than assuming a value.

Protocol. Every campaign has a pre-registration frozen with its SHA before implementation, a deviations file, and per-unit reporting. **Results are never reported as a mean alone.** Bimodality across seeds is a measured property of this bench, and §5.1 is a direct consequence of it.

Use of a language model. All experiment code, training campaigns and report generation in this work were carried out with the assistance of a large language model. The study design, the pre-registrations and their frozen hashes, the choice of controls and the verification of every reported number are the author's own. The model is not an author and bears no accountability for the work.

4 The objective

Write ℓ for the value logits, y for the label, a for the head's scalar. The self-referential target is

$$b \;=\; \mathbb{1}\big[\mathrm{sg}\big(\arg\max_{v \neq \texttt{NOSE}} \ell_v\big) \neq y\big]\,, \qquad \mathcal{L} \;=\; \mathrm{BCE}(a, b) \;+\; \mathrm{CE}_{\text{value}},$$

with sg the stop-gradient. On questions with no answer, $b = 1$ by construction. The value cross-entropy is computed only where an answer exists and normalized by that fraction, so that the loss scale does not depend on p_{nose}.

Two properties follow directly and are worth stating before any measurement, because they are what the results confirm.

The value gradient is never gated by the head. $\mathrm{CE}_{\text{value}}$ does not depend on a. Whatever happens to the head, retrieval keeps receiving gradient. The collapse reported below is therefore *not* an instance of "the model stops learning because it stopped answering"—a hypothesis we held and disproved (§5.1).

If the head ignores its input, its optimum is the prior. A constant a minimizing BCE against b is logit $\mathbb{E}[b]$, and $\mathbb{E}[b]$ is computable in closed form from RECUP:

$$\mathbb{E}[b] \;=\; p_{\text{nose}} + (1 - p_{\text{nose}})\,(1 - \text{RECUP}), \qquad p_{\text{nose}} = 0.4065. \tag{1}$$

This is a prediction with no free parameters, and §5.2 tests it.

5 Results

Nine units, seeds 0–8, identical flags. Five reach a useful state; four do not. We report per unit throughout.

A confound in the starting point, found late and reported in full. The nine units are *not* homogeneous, and we discovered this after the analysis below was written. The training harness seeds the abstention phase from a base checkpoint when one exists for that seed. Only three such bases existed (`n3_s0`, `n3_s1`, `n3_s2`, each carrying 12,000 steps of retrieval-only training), so seeds 0–2 entered the abstention phase already retrieving at RECUP 0.7734, 0.7970 and 0.7725, while seeds 3–8 started from scratch. The campaign's pre-registration asserted that only the seed varied; the check performed at the time compared the saved *configs*, which do not record whether seeding occurred.

group	seeds	starting RECUP	outcome
pre-trained	0, 1, 2	≈ 0.78	**3 of 3 useful**
from scratch	3–8	0	2 useful, **4 degenerate**

We state plainly what this costs and what it does not. **It costs the rate.** "Four of nine seeds" is not a rate for a homogeneous population; the honest statement is 0 of 3 with pre-training and 4 of 6 without. **It does not cost the phenomenon**, which is measured per unit and does not depend on the grouping: the degenerate fixed point exists, four units are in it, their heads sit at the prior, and the state is absorbing.

And it *strengthens* the mechanism of §4 rather than weakening it. If the outcome is a race between retrieval learning and head settling, then a model that arrives already retrieving has won the race before it starts—which is exactly the pattern observed. That two of the six from-scratch units reach the useful point anyway shows pre-training is not necessary, only strongly favourable.

Where this does bite, and where it does not, is spelled out in the two affected sections: §5.4 rests on a paired control that shares both seed *and* base and is therefore unaffected, while the wider 31-versus-9 tabulation is confounded and we do not rely on it.

5.1 Four of nine units end mute

What the standard metrics say about the bold rows. All four abstain on every question. Therefore `nose` = 1.0000: they correctly decline every unanswerable question. And `invento` = 0.0000: they never name content absent from the archive, because they never name anything. On the two numbers the abstention literature reports, these are the best units in the table—strictly better than `b3_s0`, which answers and therefore occasionally errs.

Their global accuracy is 0.4065: the trivial floor, to four decimals, in all four.

A hypothesis we held and disproved. `vigente` is composite, so `vigente` = 0.0000 is compatible with *perfect* retrieval hidden behind a stuck head. We expected that. It is false: RECUP is 0.30–0.40 in the degenerate units against 0.73–1.00 in the rest. Retrieval is genuinely impaired.

Table 1: The nine units at their own operating point. RECUP is retrieval ignoring the head; the last three columns are the head's behaviour. Degenerate units in bold.

unit	step	RECUP	AUC(a)	$a > 0$	median a	range of a
b3_s0	26000	0.9996	0.9997	0.407	−7.634	29.17
b3_s1	26000	0.9992	0.9999	0.407	−7.563	29.86
b3_s4	19000	0.7984	0.8084	0.413	−0.157	14.17
b3_s2	26000	0.7946	0.8163	0.392	−0.233	14.23
b3_s5	6000	0.7314	0.7550	0.529	+0.069	5.66
b3_s6	26000	**0.3964**	**0.5734**	**1.000**	+1.006	**3.12**
b3_s3	26000	**0.3820**	**0.5784**	**1.000**	+1.045	**2.83**
b3_s7	13500	**0.3432**	**0.5220**	**1.000**	+1.251	**1.06**
b3_s8	8000	**0.3040**	**0.5458**	**1.000**	+1.507	**0.85**

But it is not zero, and that matters for the interpretation. At ≈ 0.35 against a chance rate below 0.01, these models have learned a substantial part of the task and stopped short. The degenerate state is not an untrained model; it is a partially trained model whose head correctly reports that partial training as insufficient.

5.2 The head has collapsed to the prior, and the constant is where theory puts it

The four degenerate heads do not discriminate: AUC on their own target is 0.522–0.578, against 0.9997 for `b3_s0`. Nor are they saturated at $+\infty$: the entire 1–99 percentile range of their logits spans 0.85–3.24, against 29.17. They are constants sitting just above the decision threshold.

Equation (1) predicts *which* constant, with no fitted parameter:

Table 2: Closed-form prior against measured median logit. No parameters are fitted.

unit	RECUP	$\mathbb{E}[b]$	predicted logit	measured median a	difference
b3_s8	0.3040	0.8196	1.513	1.507	**0.006**
b3_s7	0.3432	0.7963	1.363	1.251	0.112
b3_s3	0.3820	0.7733	1.227	1.045	0.182
b3_s6	0.3964	0.7647	1.179	1.006	0.173

The ordering is reproduced in all four and the youngest unit falls within 0.006 of the parameter-free prediction. The older units sit slightly below their prior, which is what a head beginning to discriminate would do; consistent with this, the range of a grows monotonically with step across the four ($0.85 \to 1.06 \to 2.83 \to 3.12$). **The head is not broken. It is optimal for a generator that is wrong** 80% **of the time.**

5.3 The bifurcation is decided at step 2500

Every unit with this target passes through an initial mute phase, which is expected: at initialization the target is the constant 1. What differs is whether it ends.

The training loop already evaluates on 8 batches of 64, i.e. 512 samples. At step 2500:

The separation is zero versus non-zero, and `b3_s4` survives on a **single sample out of** 512 and reaches `vigente` = 0.68. Breaking the silence once appears to be the event that matters, which is what the race in §4 predicts: one sample on which the `argmax` is right is one sample whose target is 0, and that is what a constant head has no gradient to produce.

Table 3: Answers emitted out of 512 at step 2500, against the eventual outcome.

unit	`abstencion` @2500	answers emitted	outcome
b3_s2	0.9844	8	useful
b3_s0	0.9922	4	useful
b3_s5	0.9922	4	useful
b3_s1	0.9961	2	useful
b3_s4	0.9980	**1**	useful
b3_s3	1.0000	**0**	degenerate
b3_s6	1.0000	**0**	degenerate
b3_s7	1.0000	**0**	degenerate
b3_s8	1.0000	**0**	degenerate

Validation on the repository. Applied to **every** run in this repository with a binary abstention head that reached $\geq$ 6000 steps and has a step-2500 checkpoint—40 runs, twelve families, two decision rules, both targets—the rule separates **40 of 40**: 36 predicted useful and useful, 4 predicted degenerate and degenerate.

The hardest available test, run after the rule was formulated. The rule was derived post-hoc. We then took the two *oldest* degenerate units to full horizon under the original flags, pre-registering the prediction that neither would leave total abstention. Neither did: `b3_s3` and `b3_s6` reached step 26000 with `abstencion` $\geq$ 0.999 at every logged milestone. This is not a prospective validation—the properly prospective test would have been a screening design we did not run—but it is the strongest test the existing units permitted, and the rule survived it.

A caveat on the margin. One sample in 512 is a margin of 0.2%. As an operational criterion the abstention *rate* is too fragile; the robust form of the same quantity is $\min_i a_i$ over a large batch, which is continuous and independent of batch size.

5.4 The mute phase belongs to the self-referential target

This is the section the seeding confound touches most, so we state the limit first. Every run with a fixed target in this repository uses seeds 0–2, and those are exactly the seeds for which a pre-trained base exists. The wide tabulation below is therefore confounded with initialization, and we report it as description rather than as evidence:

target	runs	`abstencion` at step 2500
fixed ("does the answer exist?")	31	0.02–0.32
self-referential ("will I be wrong?")	9	**0.98–1.00**

The claim rests instead on the paired control, which the confound does not reach. The three `p3` units share seed, architecture, budget *and* pre-trained base with `b3_s0/s1/s2`, and differ only in the target:

seed	`abstencion` @2500, fixed target	`abstencion` @2500, self-referential
0	0.086	**0.992**
1	0.201	**0.996**
2	0.063	**0.984**

Three of three, with the confounding variable held fixed by construction. The self-referential target drives the model into the mute phase *even when it arrives already retrieving* at RECUP ≈ 0.78; the fixed target never does. The `p3` units end at RECUP 0.80–0.97 and none falls in the degenerate band.

The mechanism is the one §4 names: a fixed target is observable from step 1 and discriminative immediately, so its head never passes through the constant phase in which the degenerate solution is available.

5.5 The degenerate point is absorbing, not slow

Whether the state is a fixed point or a slow bottleneck changes what follows from it entirely, and the two are distinguishable by measuring retrieval *inside one unit*—which this line of work had never done, because intermediate checkpoints are overwritten and the logged `vigente` is 0.0000 at every milestone regardless of retrieval.

We archived every partial checkpoint of `b3_s3` from 22000 to 26000 and measured RECUP on each with paired samples ($n = 8000$, seed 54321), under a pre-registration frozen beforehand that asked for $\geq +0.0100$ and monotonicity in ≥ 3 of 4 intervals:

step	22000	23000	24000	25000	26000
RECUP	0.3841	0.3778	0.3844	0.3812	0.3820

The total is $-$**0.0021** over 4000 steps. With $p_{\text{flip}} = 0.1526$ measured, the paired standard deviation is 0.0057, so the total is 0.4σ and each interval is 0.1–1.2σ: **noise around a flat line**. Monotonicity was 2 of 4. The pre-registration's risk criterion fired and its abandonment clause applied; a planned follow-up campaign of 70000 steps was cancelled.

What this does not separate, declared in advance. The final 4000 steps run with the learning rate at the floor of its cosine schedule. "The attractor is absorbing" and "nothing moves at floor learning rate" are both consistent with a flat curve, and the cancelled campaign was what would have separated them. We therefore state the result as *absorbing under this budget and this schedule*, not as absorbing in general. This is the principal open question left by the paper.

One interval points the other way. `b3_s6`, which had only 1000 steps remaining, gives +0.0080 (1.6σ). The pre-registration had already ruled that a single interval does not constitute a trend, and it does not, but it is the one measurement in this paper pushing in the opposite direction and we report it as such.

5.6 A correction to our own first reading

Our first analysis fitted a line through the four degenerate units ordered by step and extrapolated that RECUP would reach 0.70 at ≈ 84000 steps—i.e. that the state was slow rather than absorbing. Two checks were run before publishing that reading, and the second overturned it.

The check that passed. The initial measurement used $n = 2000$, whose $\sigma = 0.0135$ is comparable to the between-unit spread, so we suspected the trend was measurement noise. Remeasuring all four at $n = 8000$ gives $r = 0.9864$ with a range of 12.5σ. **The trend is real and the suspicion was wrong.**

The check that failed. The trend is real but not linear. Per-interval slopes decrease monotonically—$+0.0071$, $+0.0048$, $+0.0014$ per 1000 steps between successive units, and -0.0005 within `b3_s3`—and a saturating fit gives residual sum of squares 12.6× smaller than the linear one. The extrapolation was an artifact of fitting a straight line to a concave curve.

What we do *not* conclude from the saturating fit. It has three parameters for four points, and its own prediction at step 26000 (0.4535) misses the measured value (0.3820) by 0.07. It is not evidence. The claim in §5.5 rests on the within-unit measurement, which has no between-seed confound, and the saturating fit only explains why the earlier extrapolation was wrong.

6 What we do not claim

This is not evidence that the self-referential target is useless. Five of nine units reach a useful state, two of them with RECUP ≥ 0.999 and AUC ≥ 0.9999. The claim is that the objective admits a second stable solution and that a substantial minority of seeds find it—not that the objective fails.

The degenerate units are not miscalibrated. It would be convenient to call them broken. They are not: their heads are optimal for their generators, to within 0.006 of the parameter-free prediction in the cleanest case. The failure is that being well calibrated about knowing nothing is worth nothing, which is a statement about the objective and the metrics, not about calibration.

We do not claim the model knows when it does not know. Everything here is supervised: the head is trained against labels, and in the self-referential case against labels derived from supervised comparison. The degenerate units in particular are a caution against that reading, since they exhibit the outward signature of knowing when they do not know while knowing nothing at all.

The step-2500 rule is post-hoc. It was formulated after observing outcomes, validated on 40 existing runs and tested on two units taken to full horizon under pre-registration. That is strong retrospective evidence and one hard confirmation; it is not a prospective validation, and we do not report it as one.

7 Limitations

- **The starting point was not controlled, and we found it late.** Seeds 0–2 inherited 12,000 steps of retrieval pre-training and seeds 3–8 did not, so the headline split is not a rate over a homogeneous population (§5.1). The per-unit phenomena are unaffected and the paired control of §5.4 is clean, but any future campaign on this question must fix the initialization across all units, and the harness must record whether seeding occurred—the saved config does not, which is why the original check missed it.
- **Scale.** 863,730 parameters, a synthetic 242-token language, one difficulty level, p_{nose} fixed at 0.4. Nothing here transfers automatically. We note only that the degenerate fixed point is a property of the objective's algebra rather than of the model's size, which makes the transfer question worth asking rather than answered.
- **Budget and schedule are confounded with the absorbing claim**, as declared in §5.5. This is the single largest gap in the paper.

- **Nine units.** The 5–4 split is a count, not a rate estimate; we report it as observed and do not attach a confidence interval to a proportion from nine seeds.
- **Two units are incomplete.** `b3_s7` and `b3_s8` were stopped at 13500 and 8000 steps after the campaign's blocking criterion became arithmetically unreachable. Their degenerate status at those steps is measured, not extrapolated, but they did not run to horizon, and the deviation is declared in the repository.
- **One decision rule.** All units use the separate binary head. Whether the token or slot interfaces admit the same fixed point under a self-referential target is untested.
- **Directed literature search, not a systematic review.**

8 Conclusion

A hallucination detector wants to predict its own model's errors, so the natural target is self-referential. That target has two stable fixed points, and the degenerate one is not a pathology to be engineered away but a correct solution: a model that retrieves almost nothing and reports, accurately, that it retrieves almost nothing. Four of the six units that began from scratch found it; none of the three that began already retrieving did.

What makes this more than a curiosity is that the degenerate solution is **invisible to the metrics the field reports**. It scores `nose` $= 1.0000$ and `invento` $= 0.0000$, better than the units that work, while its global accuracy sits at the trivial floor to four decimals. Any criterion built from abstention rate and fabrication rate is maximized by it. The remedy is not subtle—report global accuracy against its trivial floor, and require it alongside any abstention metric—but it has to be applied, and in this project it was applied only after a campaign had already been designed around criteria that a mute model would have won.

The bifurcation is legible early and cheaply. Zero answers emitted at step 2500 separated 40 of 40 runs, and one unit survived on one sample out of 512. If that rule generalizes—which this paper does not establish—then the relevant question for a self-referential abstention objective is not how to make a model abstain well. Abstaining well is trivial, and a model can do it by knowing nothing. The question is how to leave silence without beginning to invent, and that is decided in the first few thousand steps.

Data and code availability

All pre-registrations with their SHA hashes, deviation files, analysis scripts, per-seed raw results and machine-generated reports are archived in the project repository, `https://github.com/SperanzaMax/delta-archivo`, whose commit history carries the timestamps that establish the order of pre-registration and result.

Declarations

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